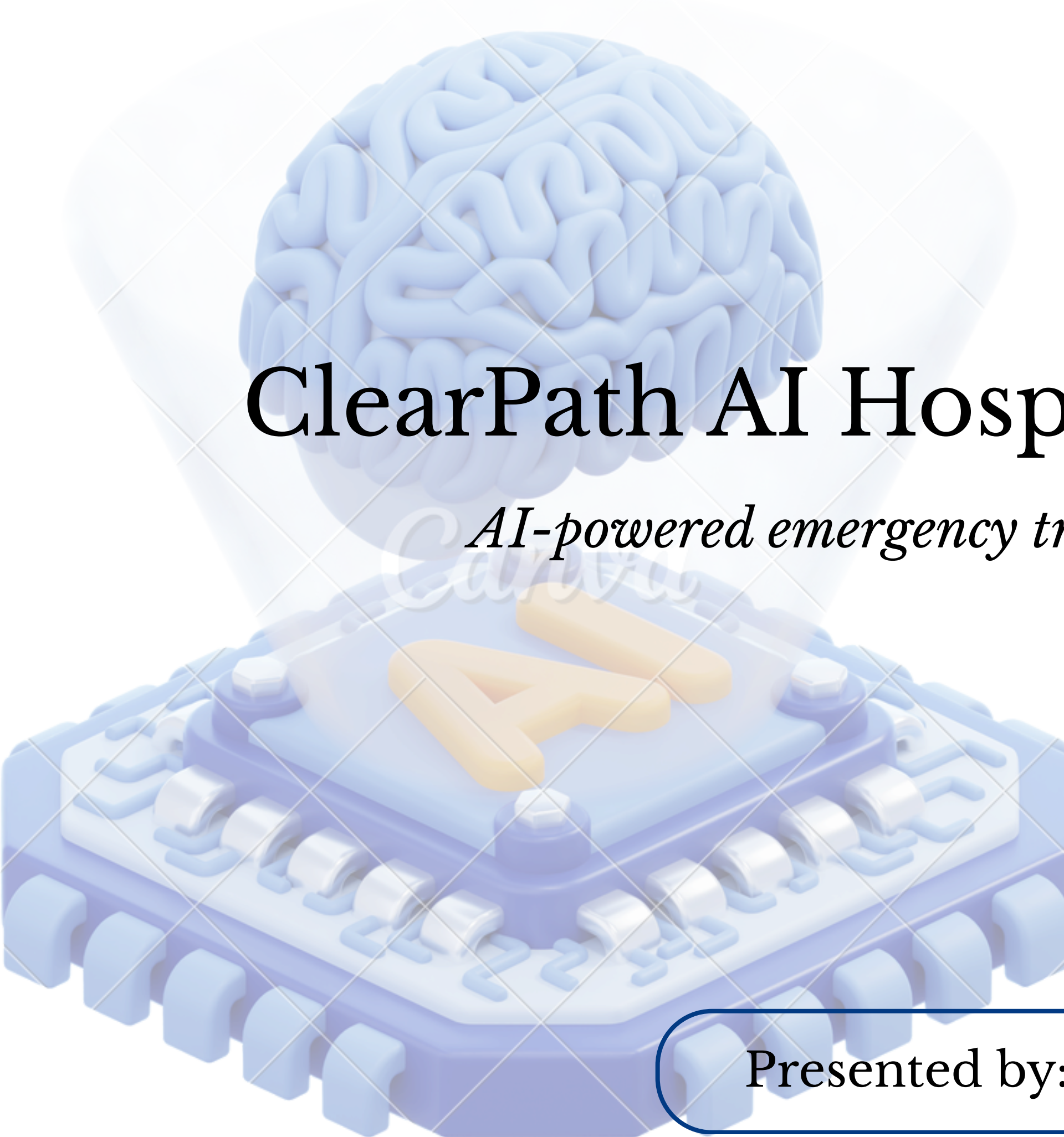




ClearPath AI Hospital Triage System

AI-powered emergency triage in under 5 seconds

Presented by: Technologia



ABOUT OUR TEAM



FU LY NING
Year 2 Engineering Student



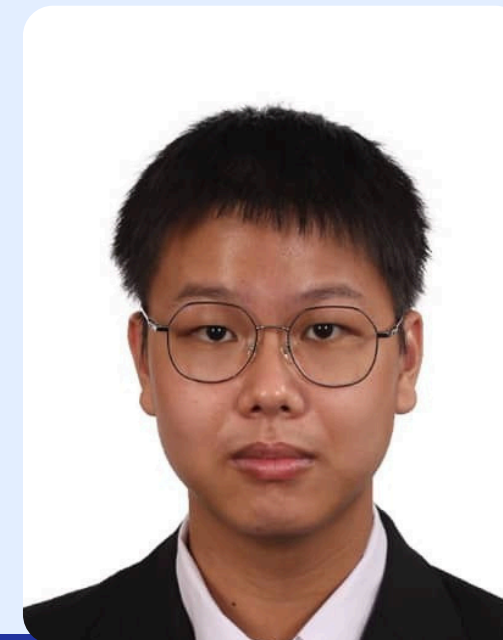
HEE ZHI DING
Year 2 Computer Science
Student



NG MUN SHEN
Year 2 Engineering Student



LAM YEAN CHIN
Year 2 Computer Science
Student



TANG JAC TZE
Year 2 Engineering Student



AN ANALYSIS OF UNDERTRIAGE AT HOSPITAL SULTAN ABDUL HALIM, SUNGAI PETANI, KEDAH, MALAYSIA

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ABSTRACT

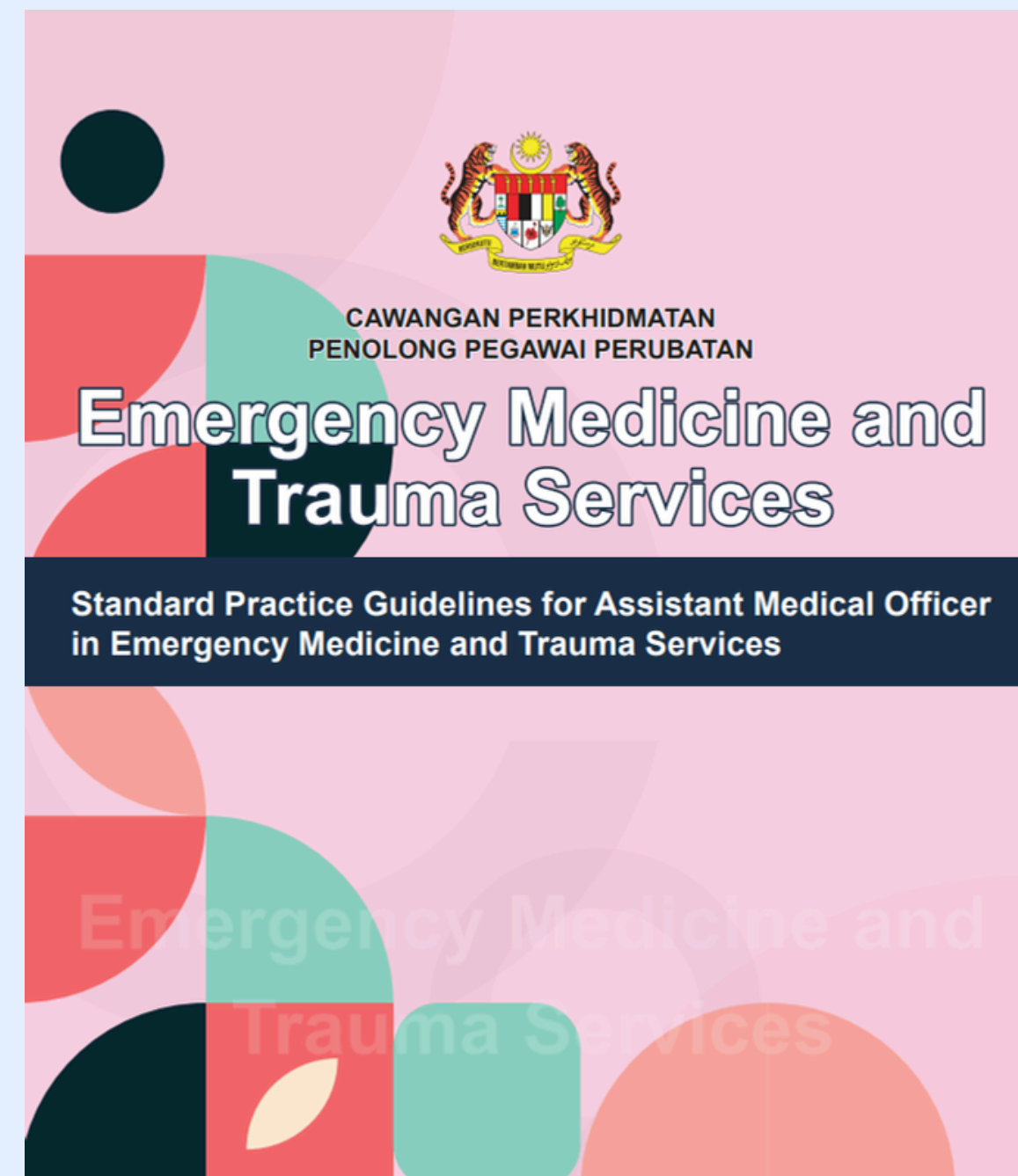
Undertriage may lead to serious outcomes such as mortality, morbidity, and medical litigations. This study aimed to determine the rate, outcomes, and predictive factors of undertriage among red zone patients at Hospital Sultan Abdul Halim, Kedah. A retrospective study was conducted from January 2018 until December 2019 involving 393 patients, with 131 and 262 patients categorized as undertriaged and appropriately triaged, respectively. Undertriage patients were identified as patients who fulfilled the red zone category based on the Malaysian Triage Category but were mistriaged to green. The numerator was the number of red zone patients who were mistriaged to green, and the denominator consisted of all red zone patients. Patient characteristics (demography, comorbidities, chief complaints, vital signs, Glasgow Coma Scale), environmental factors (working shift, weekend, public holiday), and staff factors (paramedic and nurses' working experience) were obtained from electronic medical records and proforma forms. Logistic regression was employed to analyse the data. The rate of undertriage was 1.2% (131/10,867 cases), slightly higher than the 0.5% set by the Health Ministry of Malaysia. The predictors of undertriage were comorbidity of asthma (AOR= 3.791; $p=0.008$), Glasgow Coma Scale (AOR=2.110; $p=0.002$), and patients with specific chief complaints of cardiovascular (AOR=0.189; p value <0.001), respiratory (AOR = 0.262; p value <0.0001), neurological (AOR = 0.081; p value <0.0001) and trauma (AOR = 0.129; p value <0.0001). Asthmatic and higher GCS patients were more likely to be undertriaged, while patients with specific chief complaints were less likely to be undertriaged.

Keywords: Undertriage, triage, emergency department, mistriage

INTRODUCTION

The emergency medical care system in Malaysia

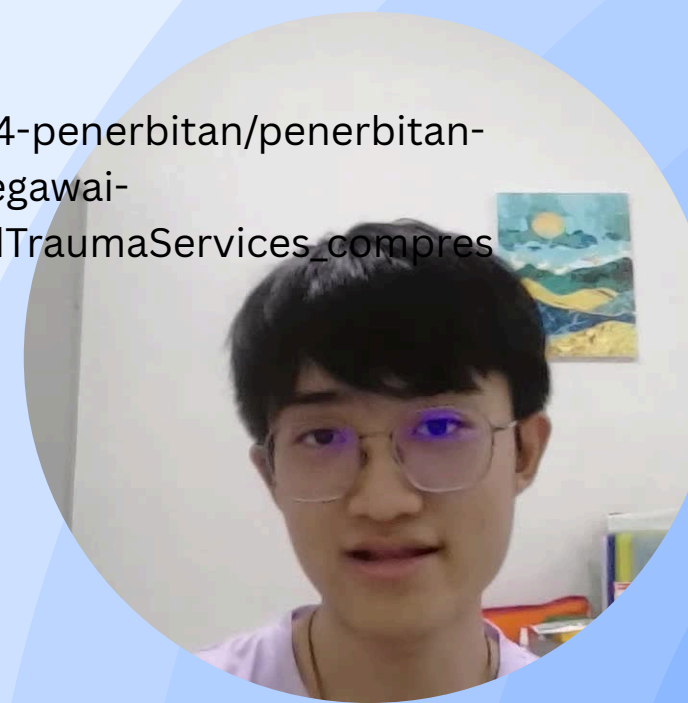
Ministry of Health (MOH) has set a standard undertriage rate of $<0.5\%$ regardless of whether the case is related to trauma or illness. A recent



Source:

https://www.moh.gov.my/images/04-penerbitan/penerbitan-klinikal/garis-panduan-penolong-pegawai-perubatan/EmergencyMedicineandTraumaServices_compressed_compressed.pdf

Source: <https://mjphm.org/index.php/mjphm/article/download/1685/619/16303>



PROBLEM STATEMENT



Triage mistakes = life-threatening

- Manual triage is slow (takes 20–45 minutes)
- Human errors & miscommunication
- Language barrier in Malaysia hospitals (BM, English, Manglish)
- No system supports multilingual clinical reasoning

Why it Matters



High patient load overwhelms ED
Delays → critical outcomes affected



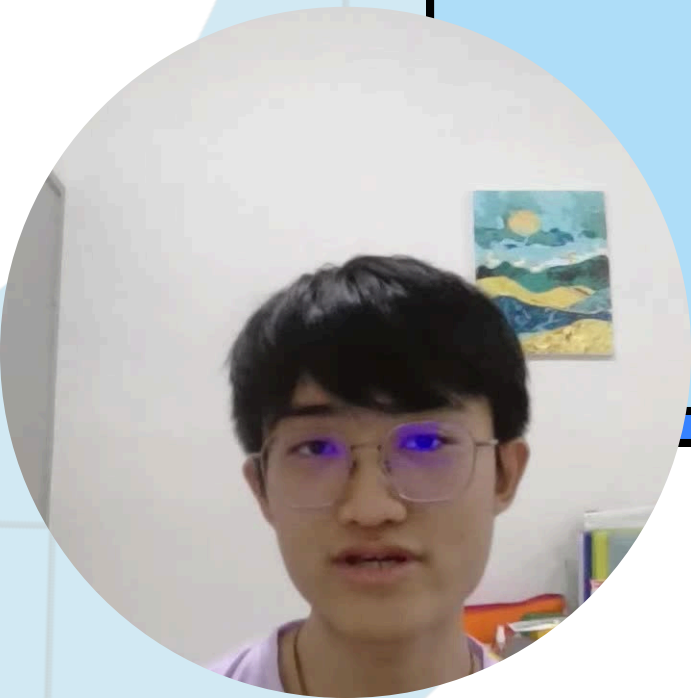
TARGET USER AND USER STORY

Triage Nurse

“As a Triage Nurse, I want to quickly classify patient severity so that critical cases are prioritized immediately”

Doctor

“As a doctor, I want automated workflow tasks generated so I can focus on treatment”



OUR SOLUTION

- AI-powered triage using GLM
- Accepts free-text multilingual input
- Outputs:
 - a) P1-P5 triage level
 - b) Clinical reasoning
 - c) Automated workflow actions
- Fully real-time system



KEY INNOVATION

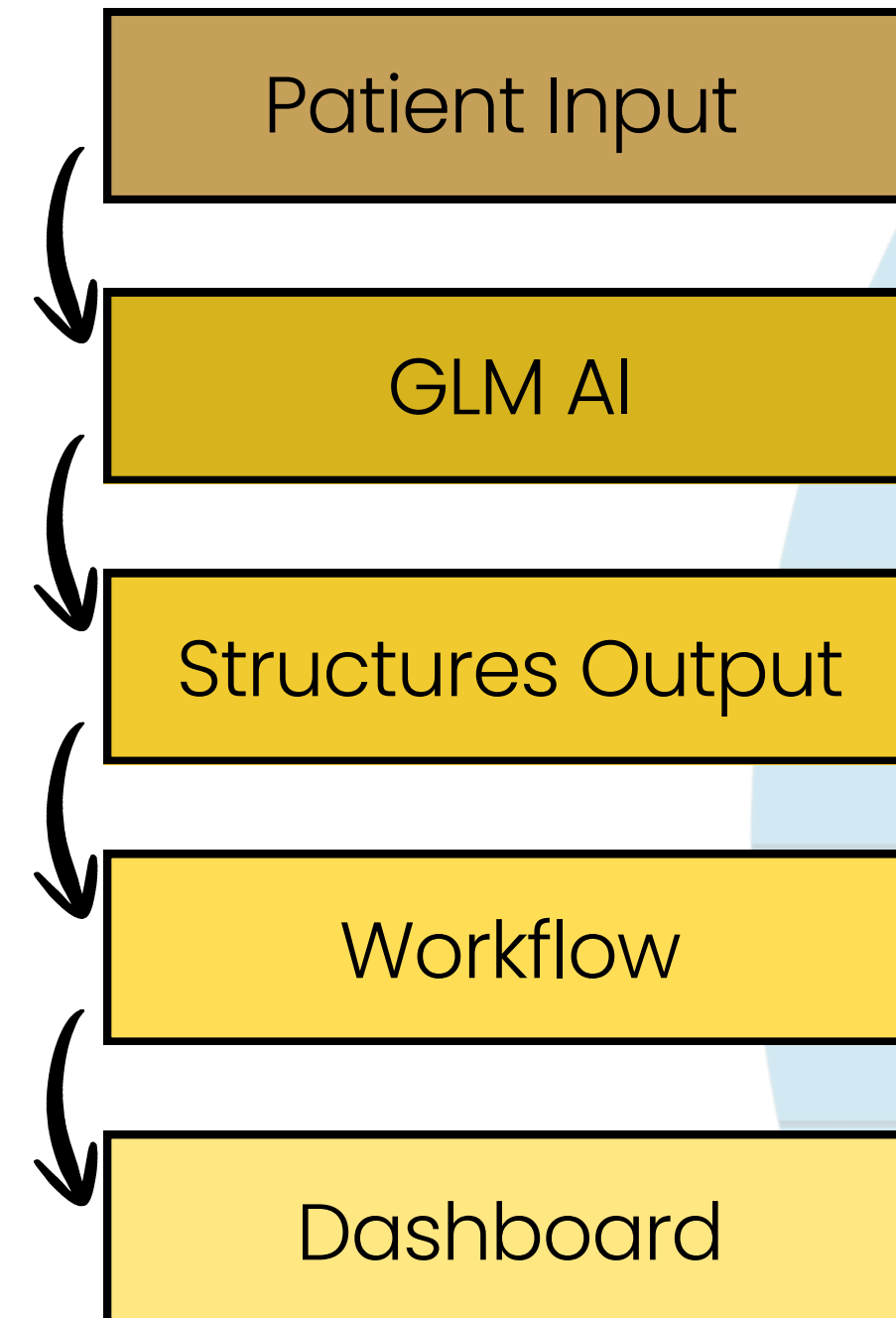
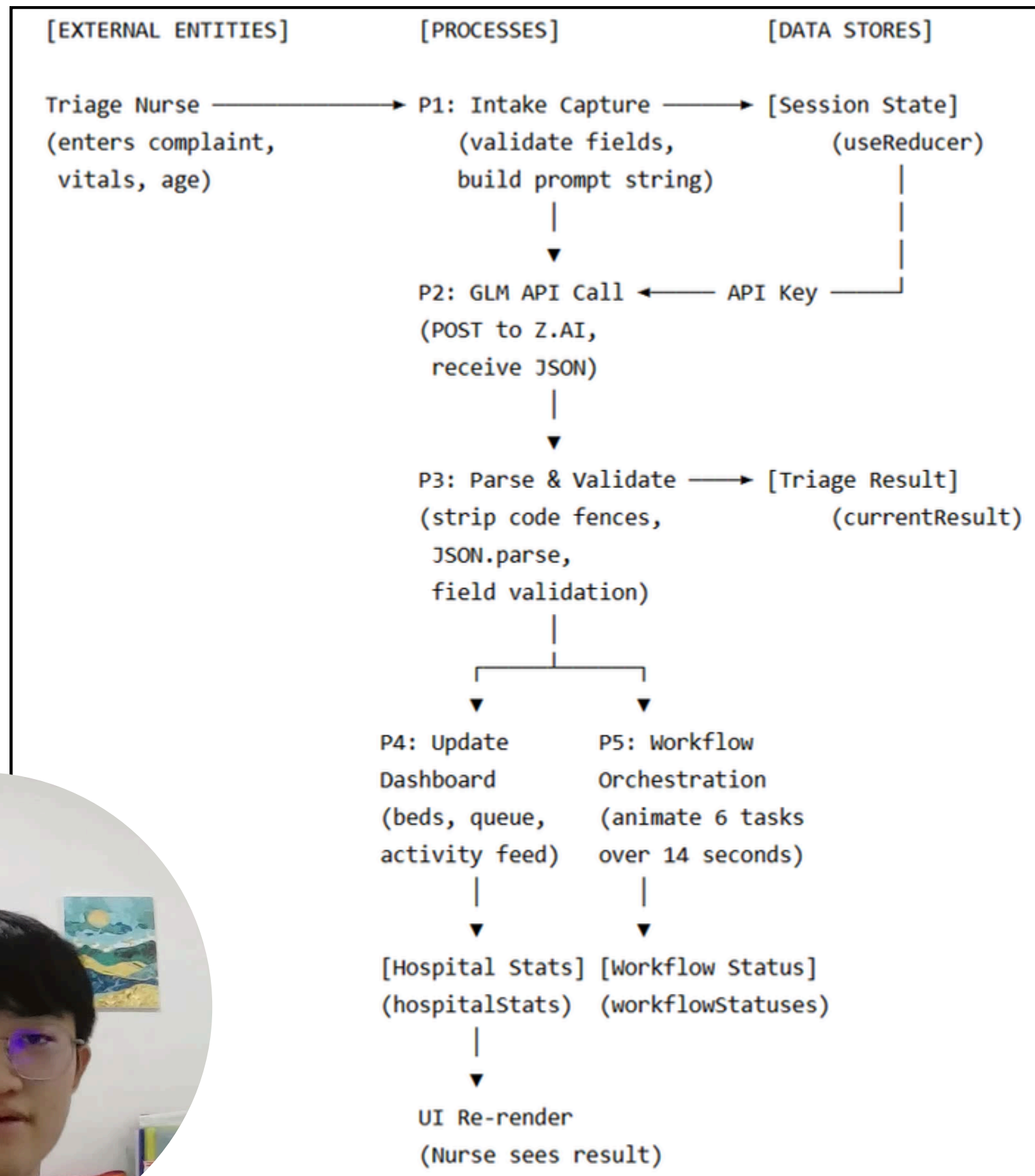
First Manglish + BM + English triage AI

End-to-end automation
(not just classification)

AI → triggers 6 hospital workflows

Designed as decision support, not
replacement

SYSTEM FLOW



Highlight:

- Unstructured → Structured transformation
- Single API call drives everything



Patient Intake (multilingual)

AI Triage Classification
(P1-P5)

Workflow Orchestration
(6 tasks)

CORE FEATURES

Live Hospital Dashboard

Demo Mode
(auto scenarios)



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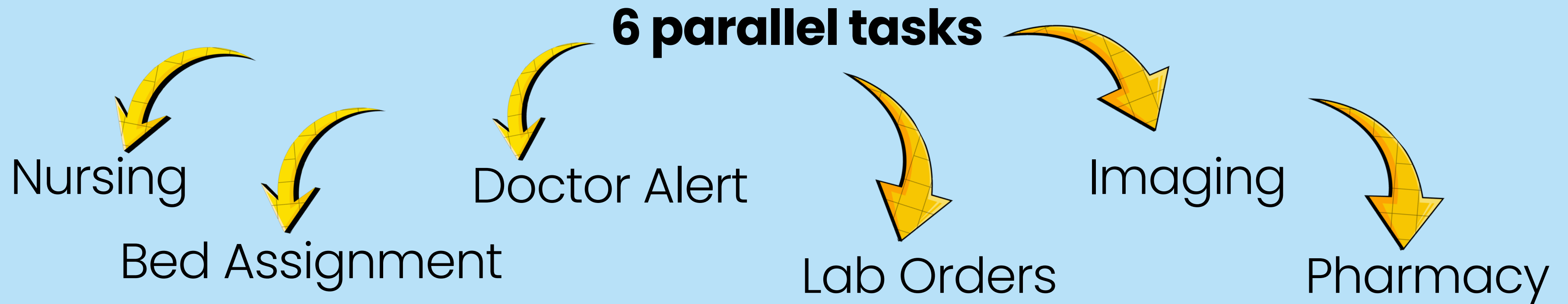
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WORKFLOW AUTOMATION



Status flow:

PENDING



IN PROGRESS



DONE

Z AI MODEL IN ACTION

a) Handles:

- Mixed language input
- Missing data (flags ambiguity)
- Clinical reasoning

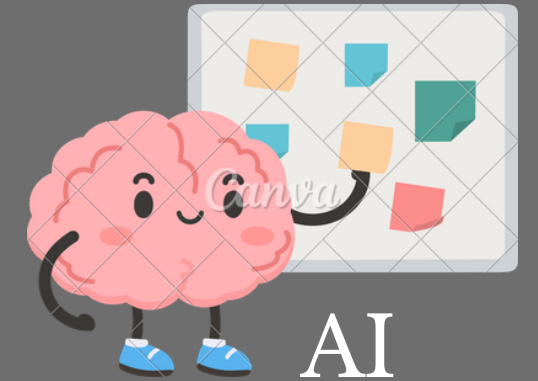
b) Output:

- Red flags
- Confidence level

c) Response time: ~2.6 seconds

PATIENT

**My chest sakit sangat,
dah 2 jam, shortness of
breath, sweating**



**Manglish, Age: 55, BP:
90/60, HR: 112, SpO2: 94**

**Red flags: chest pain, hypotension,
tachycardia, SpO2 94%**
Bed: resus/immediate
Doctor: Cardiologist STAT
Lab: ECG, Troponin, FBC
**Pharmacy: Aspirin 300mg (allergy checked),
GTN**

Confidence: HIGH

DEMO MODE

One-click auto-run of 5 pre-built clinical scenarios

P1 cardiac emergency

P2 trauma

P3 infection

P4 minor injury

P5 mild case

Fully automated pipeline

SYSTEM ARCHITECTURE

React SPA (no backend)

GLM API (AI reasoning)

Vite proxy (API handling)

State managed via useReducer

Browser-based execution



DATA & SAFETY



a) No data stored (session only)

b) API key:

- Runtime only
- Not stored

c) AI constraints:

- Strict JSON schema
- Validation checks

TESTING STRATEGY

- Unit Testing (core functions)
- Integration Testing (full pipeline)
- AI Output Testing (real scenarios)
- Edge Case Testing (errors, missing data)



RISK & RELIABILITY

- API timeout handling (30s cancel)
- Malformed AI response protection
- Missing data detection
- No system crash (graceful errors)



PERFORMANCE

- a) Avg response: ~2.6s
- b) Success rate:
 - 87.5% test pass
 - 83% AI accuracy
- c) Handles:
 - Large inputs
 - Sequential loads (Demo Mode)

IMPACT

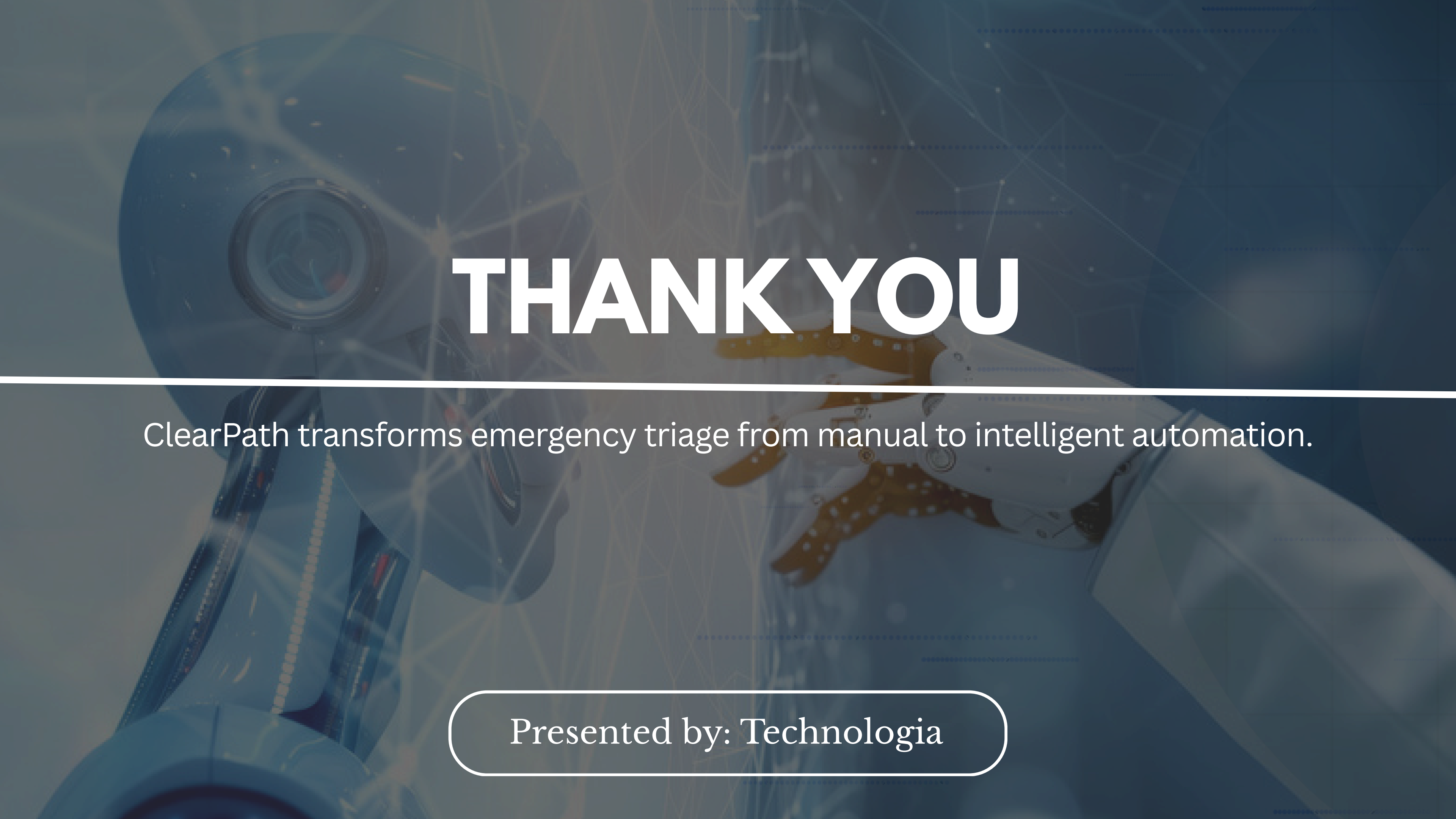
- Faster triage → save lives
- Reduce nurse workload
- Standardize decision-making
- Improve hospital efficiency

LIMITATIONS

- No real hospital integration yet
- No database (session-only)
- No multi-user system
- AI is support tool, not final decision

FUTURE IMPROVEMENTS

- HIS / EHR integration
- Multi-user dashboard
- Backend database
- Mobile/tablet support
- AI model optimization



THANK YOU

ClearPath transforms emergency triage from manual to intelligent automation.

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